

Rail Engineering Mathematics

Tutorial Sheet 4

Questions

1. Given the following complex numbers in Cartesian form:

$$z_1 = 2 + j4, \quad z_2 = 3 - j, \quad z_3 = -7 + j5, \quad z_4 = 9 - j6$$

Calculate:

(a) $z_1 + z_2$

(g) $\bar{z}_3$

(b) $z_1 - z_2$

(h) $\frac{z_4}{z_2}$

(c) $z_2 - z_1$

(i) $\frac{z_1}{z_3}$

(d) $z_3 + z_4$

(j) $z_4 \bar{z}_4$

(e) $z_3 z_1$

(k) $\text{Re}(z_1 z_3)$

(f) $z_2 z_4$

(l) $\text{Im}(z_1 \bar{z}_1)$

2. Determine the roots of the following polynomial functions:

(a) $y = x^2 + 14x + 58$

(b) $y = 4x^2 - 12x + 10$

(c) $y = x^2 + 3x + 12$

3. In the hydrogen atom, the angular momentum p of the de Broglie wave is given by:

$$p\Psi = -\left(\frac{j\hbar}{2\pi}\right)(\pm jm\Psi)$$

Determine a simplified expression for p .